

1.Cover Page

Title: Career Progression and Promotion Gap Analysis for Retention Optimization at Palo Alto Networks

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Tools Used: Microsoft Power BI, DAX Measures, Filtering using Slicers, Data Cleaning & Analysis, Power Query

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2.Acknowledgement

I would like to express my sincere gratitude to my faculty mentor and institution for providing me with the opportunity to undertake this project, “Career Progression and Promotion Gap Analysis for Retention Optimization at Palo Alto Networks.”

This project enabled me to apply data analytics concepts and visualization techniques using Microsoft Power BI to solve a real-world human resources problem. Through this analysis, I gained valuable experience in data cleaning, feature engineering, exploratory data analysis, dashboard development, and business intelligence reporting.

I would also like to thank all the individuals whose guidance, support, and feedback contributed to the successful completion of this project.

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Pawan Lakhera

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4. Abstract

Employee retention remains one of the most critical challenges for modern organizations. Delayed career progression, promotion stagnation, and inconsistent managerial support can negatively affect employee engagement and increase turnover risk. This study analyzes workforce data from Palo Alto Networks to identify career progression patterns, promotion gaps, and retention opportunities among employees.

Using Power BI, exploratory data analysis was conducted on 1,470 employees across multiple departments. The analysis identified high-risk employee clusters, departments with

elevated promotion gaps, and employees requiring career intervention. Findings indicate that promotion delays and role stagnation significantly influence attrition risk. The study concludes with strategic recommendations aimed at improving employee retention, enhancing career development pathways, and strengthening managerial effectiveness.

5.Introduction

In highly competitive industries, retaining skilled employees has become a strategic priority. Organizations invest substantial resources in recruitment and employee development; however, ineffective promotion practices and limited career growth opportunities often contribute to voluntary turnover.

Palo Alto Networks, a global cybersecurity organization, seeks to understand the factors influencing employee retention by evaluating career progression trends, promotion gaps, and managerial stability. Through data analytics, this study aims to identify workforce segments that require intervention and provide evidence-based recommendations to optimize retention strategies.

6.Problem Statement

Despite maintaining a strong workforce, employees may experience delayed promotions, role stagnation, and limited career advancement opportunities. These challenges can increase dissatisfaction and contribute to higher attrition rates.

Palo Alto Networks requires a data-driven approach to:

- Identify employees at risk of stagnation.
- Detect departments with significant promotion gaps.
- Understand the relationship between career progression and attrition.
- Develop targeted retention strategies to reduce employee turnover.

7.Objectives

The objectives of this study are to:

- Analyze employee distribution and attrition patterns.
- Evaluate promotion gaps across departments and job roles.
- Identify high-risk career clusters.
- Assess the impact of managerial stability on employee stagnation.
- Detect employees requiring career intervention.
- Recommend retention strategies to improve workforce stability.

8.Dataset Overview

The analysis was performed using employee workforce data consisting of demographic, performance, promotion, and managerial variables.

Key variables included:

Field Name	Description
Age	Age of the employee in years
Attrition	Indicates whether the employee left the organization (0 / 1)
BusinessTravel	Frequency of business travel (Non-Travel, Travel Rarely, Travel Frequently)
DailyRate	Daily compensation rate of the employee
Department	Department where the employee works (e.g., R&D, Sales, HR)
DistanceFromHome	Distance between employee's home and workplace

Education	Education level (1:Below College, 2:College, 3:Bachelor, 4:Master, 5:Doctor)
EducationField	Field of education (e.g., Life Sciences, Medical, Technical)
EnvironmentSatisfaction	Satisfaction with the work environment (1 = Low, 4 = High)
Gender	Gender of the employee (Male / Female)
HourlyRate	Hourly wage of the employee
JobInvolvement	Level of involvement in work (1 = Low, 4 = High)
JobLevel	Job seniority level (1 = Entry level, higher values = senior roles)
JobRole	Specific job role or designation of the employee
JobSatisfaction	Satisfaction with the job role (1 = Low, 4 = High)
MaritalStatus	Marital status of the employee (Single, Married, Divorced)
MonthlyIncome	Monthly salary earned by the employee
MonthlyRate	Monthly compensation rate (administrative pay metric)
NumCompaniesWorked	Number of companies the employee has worked for previously
OverTime	Indicates whether the employee works overtime (Yes / No)
PercentSalaryHike	Percentage increase in salary during the last appraisal
PerformanceRating	Performance evaluation score of the employee
RelationshipSatisfaction	Satisfaction with workplace relationships (1 = Low, 4 = High)

StockOptionLevel	Level of stock options granted to the employee
TotalWorkingYears	Total professional experience across all companies (in years)
TrainingTimesLastYear	Number of training programs attended in the last year
WorkLifeBalance	Work-life balance rating (1 = Poor, 4 = Excellent)
YearsAtCompany	Number of years the employee has worked at Palo Alto Networks
YearsInCurrentRole	Number of years in the current role
YearsSinceLastPromotion	Number of years since the last promotion
YearsWithCurrManager	Number of years working with the current manager

Power BI was used for data modeling, KPI development, and dashboard creation.

9.EDA (Exploratory Data Analysis)

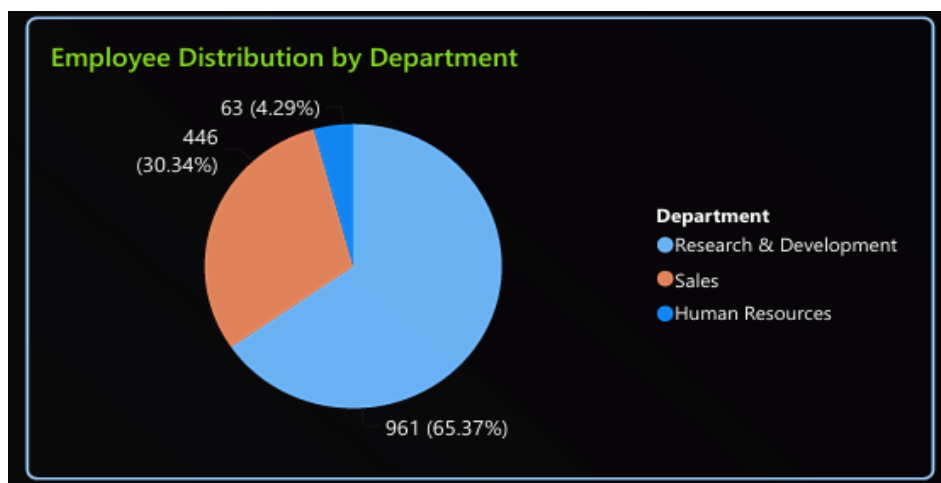
Section- A: Executive overview Dashboard

1.KPI Analysis-

Total Employees	Active Employees	Attrition Rate %	Avg Tenure	Avg Promotion Gap Ratio
1470	1233	16.12%	7.01	0.29

KPI	values	Interpretation
Total Employees	1470	Full workforce captured in the dataset.
Active Employees	1233	Employees currently with the organization
Attrition Rate %	16.12%	Roughly 1 in 6 employees has left
Avg Tenure	7.01	The average employee has spent 7 years at the company
Avg Promotion Gap Ratio	0.29	On average, employees have gone without a promotion for 0.29 of their total tenure

2.Employee Distribution by Department

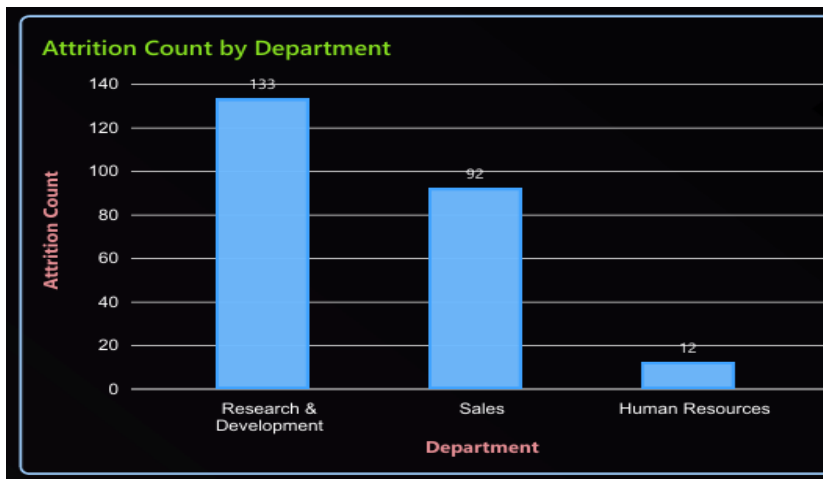


Findings-

- Research & Development is the largest department (65.37% of the workforce)

- Sales, despite being less than half the size of R&D — indicating Sales experiences proportionally greater turnover pressure.
- Human Resources, while the smallest department close to Sales.

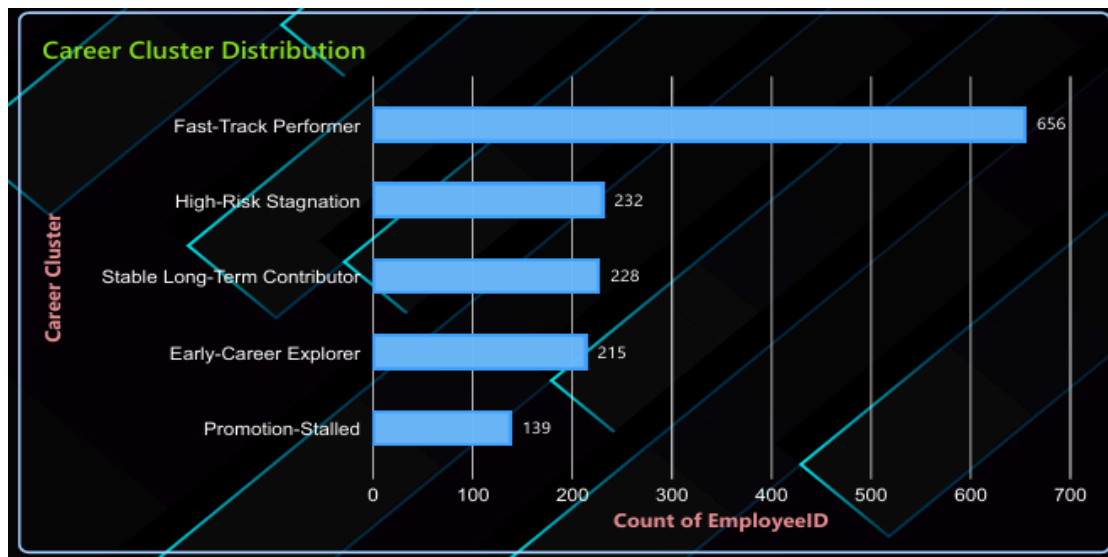
3. Attrition Count by Department



Findings-

- R&D contributes nearly half of all attrition cases.
- Sales also demonstrates considerable turnover.
- HR exhibits strong retention

4. Career Cluster Distribution

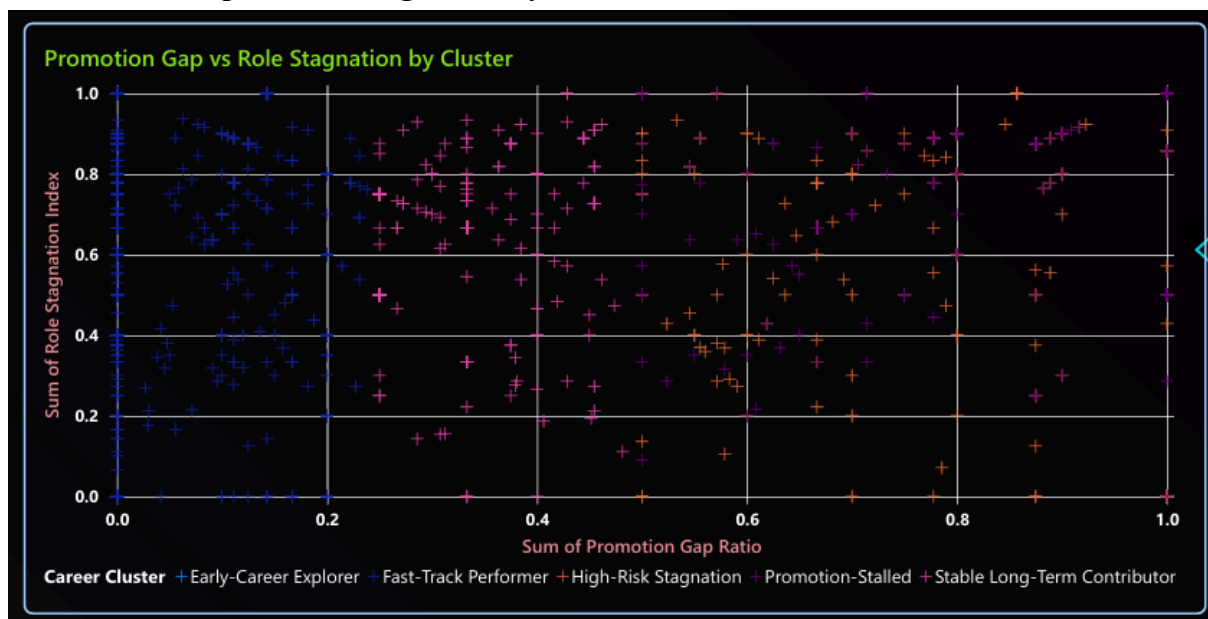


Findings-

- Fast-Track Performers dominate the workforce.
- Nearly one-quarter of employees fall into vulnerable career states (High-Risk + Promotion-Stalled).

Section- B: Career Path Clustering Analysis

1. Promotion Gap vs Role Stagnation by Cluster-



Findings-

a. Early-Career Explorers

- Mostly concentrated on the **left side (low promotion gap)**.
- Wide spread in stagnation scores.
- Employees are relatively new and have not yet accumulated promotion delays.

b. Fast-Track Performers

- Located predominantly in the **low promotion gap region (0–0.2)**.
- Moderate-to-high stagnation variation.
- Indicates employees continue progressing despite differing role experiences.

c. High-Risk Stagnation

- Clustered around **promotion gap ratios of 0.5–0.8**.
- Moderate to high stagnation levels.
- Represents employees experiencing both advancement barriers and stagnant roles.

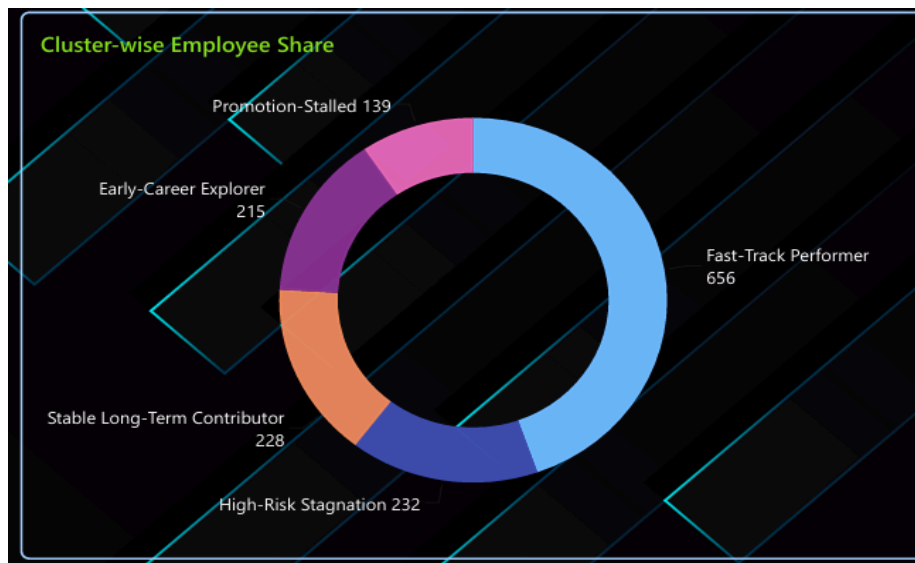
d. Promotion-Stalled

- Mostly found in the **upper-right region**.
- High promotion gap (0.7–1.0).
- High stagnation levels (0.6–1.0).

Stable Long-Term Contributors

- Distributed across moderate promotion gaps.
- Less concentrated than other clusters.
- Reflects employees maintaining stability despite slower growth.

2.Cluster-wise Employee Share



finding-

- While almost half of employees are progressing successfully, approximately one-quarter of the workforce exhibits signs of career stagnation, indicating an urgent need for targeted development and promotion interventions.

3.Cluster Profile Summary

Career Cluster Profile Summary

Career Cluster	Avg Promotion Gap Ratio	Attrition Rate %	Total Employees	Avg Performance Rating
Early-Career Explorer	0.06	34.88%	215	3.17
Fast-Track Performer	0.06	10.67%	656	3.15
High-Risk Stagnation	0.79	18.53%	232	3.16
Promotion-Stalled	0.81	16.55%	139	3.14
Stable Long-Term Contributor	0.33	11.40%	228	3.14

Findings-

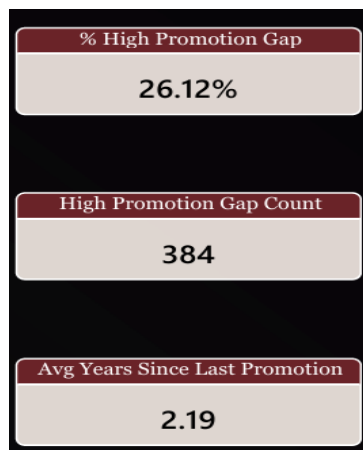
- Fast-Track Performer combines the lowest promotion gap (0.06) with the lowest attrition (10.67%), confirming that visible, rapid career movement is strongly associated with retention.
- Early-Career Explorer shows the highest attrition of any cluster (34.88%) despite an equally low promotion gap (0.06) — since this cluster captures employees with one

year of tenure or less, its high attrition reflects early-tenure churn rather than promotion stagnation.

- High-Risk Stagnation and Promotion-Stalled both show very high promotion gap ratios (0.79 and 0.81 respectively), validating that these segments correctly isolate employees experiencing genuine structural stagnation.
- Average Performance Rating is nearly identical across all five clusters (3.14–3.17) — this is the central finding of the clustering analysis: promotion stagnation is not explained by underperformance. High performers are just as likely to be promotion-stalled as low performers, which is precisely the structural blind spot traditional attrition models miss

Sec- C: Promotion Gap Monitor

1.KPI



Finding-

- **% High Promotion Gap: 26.12%** — More than one in four employees across the organisation falls into the High Promotion Gap category, indicating that promotion stagnation is a widespread structural issue rather than an isolated occurrence.

- **High Promotion Gap Count: 384** — A total of 384 employees out of 1,470 have a Promotion Gap Ratio of 0.5 or above, meaning they have gone without a promotion for at least half of their total tenure at the organisation.
- **Avg Years Since Last Promotion: 2.19 years** — On average, employees have waited over two years since their most recent promotion.

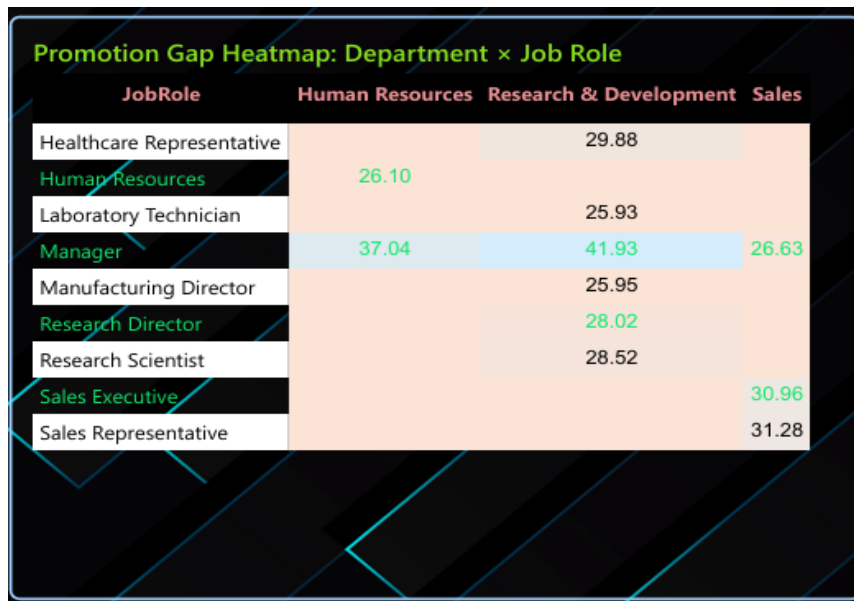
2.Average Years Since Last Promotion by Job Role



Findings-

- Managers experience the longest average promotion gap (4.8 years), nearly four times longer than Sales Representatives (1.1 years). This may reflect that Manager is already a senior role with fewer further promotion levels available, but it could equally signal a structural ceiling for high performers who have already reached management.
- Sales Representative and Human Resources roles show the shortest average gaps (1.1–1.3 years), consistent with more frequent role-level promotions in entry and early-career tracks.

3.Promotion Gap Heatmap: Department × Job Role



finding-

- The Manager role records the highest heat score in every department — 41.93 in Research & Development, 37.04 in Human Resources, and 26.63 in Sales — confirming that promotion stagnation at the Manager level is an enterprise-wide pattern rather than a department-specific issue.
- Within Sales, individual-contributor roles (Sales Representative at 31.28 and Sales Executive at 30.96) show heat scores close to the department's Manager score (26.63), indicating that promotion stagnation in Sales is distributed across the entire role hierarchy rather than being concentrated at the top.
- Within Research & Development, the heat score spread across non-managerial roles is comparatively narrow (25.93–29.88), suggesting a more consistent promotion cadence outside of the Manager role in that department.

4.High Promotion Gap Employees

EmployeeID	Department	JobRole	Sum of YearsAtCompany	Sum of YearsSinceLastPromotion	Sum of PerformanceRating
1	Sales	Sales Executive	6	0	3
2	Research & Development	Research Scientist	10	1	4
3	Research & Development	Laboratory Technician	0	0	3
4	Research & Development	Research Scientist	8	3	3
5	Research & Development	Laboratory Technician	2	2	3
6	Research & Development	Laboratory Technician	7	3	3
7	Research & Development	Laboratory Technician	1	0	4
8	Research & Development	Laboratory Technician	1	0	4
9	Research & Development	Manufacturing Director	9	1	4
10	Research & Development	Healthcare Representative	7	7	3
11	Research & Development	Laboratory Technician	5	0	3

findings-

- A significant proportion of employees experience delayed career progression.
- Promotion gaps are not necessarily linked to poor performance.
- Promotion delays are more prevalent among experienced employees.
- High promotion gaps can increase retention risk.

Sec- D: Retention Opportunity Analysis

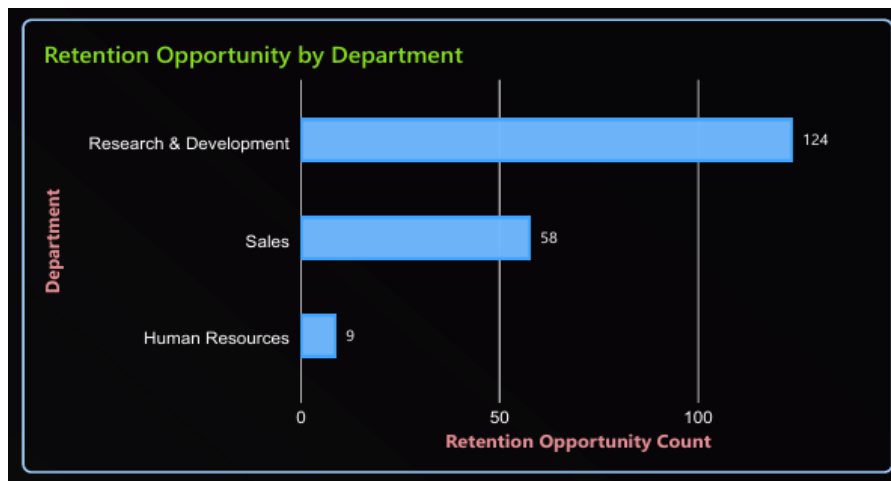
1.KPI

Retention Opportunity Rate %	Retention Opportunity Count
15.49%	191

Findings-

- **Retention Opportunity Count: 191** — A total of 191 currently active employees have been identified as retention opportunities.
- **Retention Opportunity Rate %: 15.49%** — Approximately 15 out of every 100 active employees represent a hidden retention risk that standard attrition models would not detect.

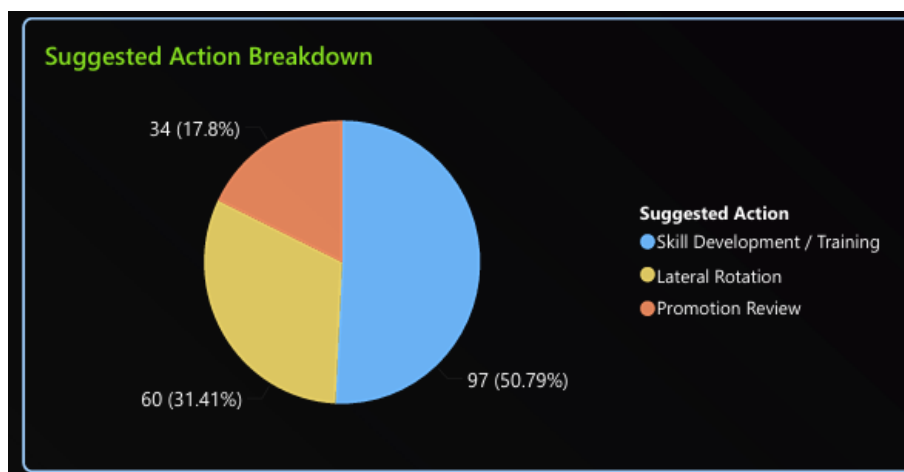
2.Retention Opportunities by Department



Finding-

- The retention opportunity rate by department is: Human Resources 9 out of 51 active employees, Sales- 58 out of 354, and Research & Development- 124 out of 828.
- Although Research & Development contributes the largest absolute number of flagged employees (124), its retention opportunity rate is the lowest of the three departments. Human Resources, despite having the fewest flagged employees in absolute terms (9), meaning it has the largest share of its active workforce sitting in this blind-spot segment.

3.Suggested Action Breakdown



Finding-

- Skill Development / Training is the most common recommended intervention, applying to just over half (50.79%) of flagged employees, followed by Lateral

Rotation (31.41%) and Promotion Review (17.8%). This indicates that for the majority of at-risk employees, addressing training intensity is a more immediate and lower-cost lever than role rotation or formal promotion review

4. Employees Flagged for Career Intervention

EmployeeID	Department	JobRole	Sum of YearsSinceLastPromotion	Sum of JobSatisfaction	Suggested Action
38	Sales	Sales Representative	2	4	Lateral Rotation
153	Sales	Sales Representative	4	3	Skill Development / Training
227	Sales	Sales Representative	5	4	Skill Development / Training
363	Sales	Sales Representative	2	4	Promotion Review
471	Sales	Sales Representative	2	4	Lateral Rotation
676	Sales	Sales Representative	5	3	Skill Development / Training
755	Sales	Sales Representative	2	4	Lateral Rotation
819	Sales	Sales Representative	2	4	Lateral Rotation
877	Sales	Sales Representative	2	4	Promotion Review
1101	Sales	Sales Representative	4	3	Promotion Review
1179	Sales	Sales Representative	2	3	Lateral Rotation
1329	Sales	Sales Representative	5	3	Lateral Rotation
1418	Sales	Sales Representative	1	3	Lateral Rotation
47	Sales	Sales Executive	8	3	Skill Development / Training
83	Sales	Sales Executive	1	4	Promotion Review
132	Sales	Sales Executive	2	4	Lateral Rotation
139	Sales	Sales Executive	2	3	Lateral Rotation
168	Sales	Sales Executive	7	4	Skill Development / Training
219	Sales	Sales Executive	12	4	Skill Development / Training
242	Sales	Sales Executive	2	4	Promotion Review
262	Sales	Sales Executive	7	4	Skill Development / Training
282	Sales	Sales Executive	11	3	Skill Development / Training
304	Sales	Sales Executive	7	4	Skill Development / Training
355	Sales	Sales Executive	2	3	Lateral Rotation
356	Sales	Sales Executive	7	3	Skill Development / Training
385	Sales	Sales Executive	9	3	Skill Development / Training
402	Sales	Sales Executive	4	2	Lateral Rotation

Finding-

A targeted group of employees requires immediate retention attention.

- A total of 191 employees were flagged for career intervention, accounting for 15.49% of active employees.
- These employees exhibit characteristics associated with elevated retention risk.

Career intervention needs extend beyond promotion concerns.

- Employees requiring intervention often demonstrate a combination of:
 - High promotion gaps,
 - Role stagnation,
 - Frequent managerial changes,
 - Moderate job satisfaction levels.

Skill development emerged as the most common intervention strategy.

- This highlights the importance of continuous learning in improving employee engagement and progression.

Not all intervention cases require promotion.

- Recommended actions also include:
 - Lateral role rotations to provide new growth opportunities.
 - Promotion reviews for employees demonstrating readiness for advancement.
- This emphasizes the need for personalized retention strategies rather than a one-size-fits-all approach.

Research & Development contributes the largest share of intervention cases.

- Due to its substantial workforce size, the Research & Development department recorded the highest number of employees requiring intervention.
- Focused retention initiatives within this department could yield significant organizational benefits.

Sec- E: Managerial Insight Analysis

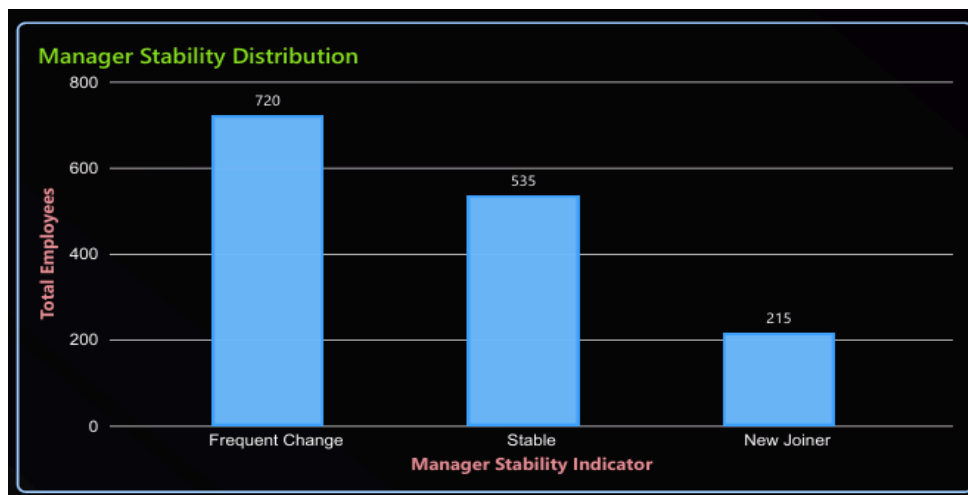
1.KPI

Avg Years With Manager	Stable Manager %
4.12	36.39%

findings-

- **Avg Years With Current Manager: 4.12 years** — On average, employees have spent just over four years working under their current manager.
- **Stable Manager %:** Only 36.39% of the workforce has a manager stability profile classified as Stable, meaning fewer than four in ten employees have spent a consistently high proportion of their tenure under the same manager. This figure highlights that a stable, continuous reporting relationship is a minority experience at Palo Alto Networks rather than the organisational norm.

2.Manager Stability Distribution



finding-

- The workforce is distributed across three manager stability categories: Frequent Change (720 employees), Stable (535 employees), and New Joiner (215 employees).
- Nearly half the workforce experiences frequent manager changes, meaning a stable, consistent reporting relationship is the minority experience rather than the norm.

- Attrition rates by manager stability- The dominant driver of high attrition is early tenure stage — not managerial instability — since New Joiners show the highest attrition rate regardless of their manager relationship.

3.Role Stagnation by Department and Manager Stability



finding-

- In every department, employees with a Stable manager relationship show the highest average Role Stagnation Index (0.63–0.75), while New Joiners show the lowest (0.00–0.10) by definition, as they have not yet had sufficient time to remain in a single role for an extended period. Frequent Change employees fall in between (0.58–0.64) across all departments.
- This is a counter-intuitive finding: a stable manager relationship does not accelerate role movement — it is actually associated with longer periods spent in the same role.
- Sales shows the widest gap between Stable (0.75) and Frequent Change (0.64) employees, suggesting that in Sales specifically, a long-standing manager relationship is most strongly associated with extended role tenure.

4. Team-Level Stability Overview

Team-Level Stability Overview			
Department	Frequent Change	New Joiner	Stable
Human Resources	35	5	23
Research & Development	466	149	346
Sales	219	61	166

Finding-

- Research & Development carries the largest absolute count of Frequent Change employees (466), which simply reflects its larger overall headcount.

10. Key Findings-

- Palo Alto Networks reported an attrition rate of 16.12%.
- Research & Development accounts for 65.37% of the workforce.
- Early-Career Explorers exhibit the highest attrition rate (34.88%).
- Approximately 26.12% of employees experience substantial promotion gaps.
- A total of 384 employees were identified with high promotion gaps.
- Nearly 15.49% of employees require retention intervention.
- Skill development was the most common recommended retention strategy.
- Frequent managerial changes were associated with higher role stagnation.

11. Recommendations-

- Prioritise retention review for Sales, given its disproportionately high attrition rate (20.63%) relative to its workforce size.
- Use Career Cluster distribution as a recurring quarterly metric to track whether the Fast-Track Performer share (currently 44.6%) is growing or shrinking over time
- Implement structured career development programs for employees identified within High-Risk Stagnation and Promotion-Stalled clusters.

- Establish transparent promotion criteria and communicate advancement pathways clearly across departments.
- Expand learning and development programs, particularly for employees flagged for intervention.
- Strengthen managerial continuity and leadership effectiveness through succession planning and manager development initiatives.
- Conduct periodic promotion gap reviews to identify employees experiencing prolonged career stagnation.

12. Conclusion

This study demonstrates that career progression significantly influences employee retention at Palo Alto Networks. Employees experiencing prolonged promotion gaps and role stagnation exhibit elevated retention risks, while managerial stability contributes positively to employee development and engagement.

By adopting proactive career management practices, strengthening development opportunities, and ensuring fair promotion processes, Palo Alto Networks can create a more resilient workforce and enhance its long-term organizational performance.